

# TEAM CONTRIBUTION & WORK DISTRIBUTION REPORT

Course: DLE-AI-202 (Deep Learning), Cohort I 2026 | Track 1: Pure Research

Project: Do Register Tokens Regularize Vision Transformers Under Data Scarcity? A Controlled 12-Run Ablation

Repository: github.com/shahin1717/vit-research | Submission Date: September 8, 2026

## 1. Executive Overview & Compliance Statement (§6)

Per Section 6 of the project specifications, this report documents the workload allocation, code ownership, and technical deliverables across our team. The project investigated the regularizing capacity of register tokens ( $K \in \{0, 1, 4, 8\}$ ) under extreme data scarcity (~10k CIFAR-100 images). All five members contributed equally (20.0% per member) across theoretical formulation, PyTorch engineering, A100 GPU sweeps, statistical reduction, and manuscript drafting.

## 2. Responsibility Matrix & Effort Allocation

| Member Name          | GitHub / Contact                             | Core Ownership Domain                | Primary Assigned Codebase Modules                                 | Effort |
|----------------------|----------------------------------------------|--------------------------------------|-------------------------------------------------------------------|--------|
| Shahin Alakparov     | shahin1717<br>sahinalekperov5@gmail.com      | Core Architecture & Training Lead    | src/models/register_vit.py, scripts/train.py, scripts/eval.py     | 20.0%  |
| Gulnisa Abdurahmanli | gulnisa<br>gulnisa.abdurahmanli@gmail.com    | Data Engineering & Pipeline Lead     | src/data/cifar100_subset.py, src/data/__init__.py                 | 20.0%  |
| Narmina Ibrahimova   | nnrmina<br>narminaibrahimova2@gmail.com      | Metrics & Interpretability Lead      | src/models/attention_hook.py, src/metrics/entropy.py, outliers.py | 20.0%  |
| Emil Ahmedli         | emilahmedli5<br>emilahmedli1905@gmail.com    | Hardware Execution & Sweeps Lead     | scripts/run_sweep.sh, configs/*.yaml, team1.conf (WireGuard)      | 20.0%  |
| Rufet Dosteliyev     | rufetdosteliyev<br>rufetdosteliyev@gmail.com | Results Engine, Figures & Paper Lead | src/utlis/export_latex.py, scripts/plot_metrics.py, paper/        | 20.0%  |
| TOTALS               | 5 Team Members                               | Balanced End-to-End Pure Research    | 12/12 Runs Verified (100% Passing Tests)                          | 100.0% |

## 3. Detailed Technical Contributions per Member

- Shahin Alakparov (20.0%):** Engineered RegisterVisionTransformer wrapping timm ViT-Tiny ( $d=192$ ). Implemented prepend logic for learnable registers  $R \in \mathbb{R}^{K \times d}$  and output slicing. Developed the AMP training harness in scripts/train.py (AdamW, linear warmup + cosine annealing, gradient clipping, checkpoint serialization) and verified 50/50 tests.
- Gulnisa Abdurahmanli (20.0%):** Built StratifiedCIFAR100Subset in src/data/cifar100\_subset.py, sampling exactly 100 samples/class (10k images) with deterministic generator seeds. Implemented 9,000 train / 1,000 val splits, Bicubic RandomResizedCrop(224), AutoAugment, and standard test set evaluation loader with pinned memory and multi-worker prefetching.
- Narmina Ibrahimova (20.0%):** Authored ViTAttentionHookManager in src/models/attention\_hook.py, non-invasively intercepting attention weights across all 12 MHSA layers with automatic CUDA memory clearing. Coded Shannon attention entropy  $H(A^{(l)})$  in src/metrics/entropy.py, patch-norm  $3\sigma$  outlier rate in src/metrics/outliers.py, and generalization gap metrics.
- Emil Ahmedli (20.0%):** Configured WireGuard VPN (team1.conf) and JupyterLab on the remote NVIDIA A100-SXM4 GPU cluster. Structured YAML profiles (configs/\*.yaml) across treatment arms and seeds {42, 1337, 3407}. Automated batch execution via scripts/run\_sweep.sh with GPU cache clearing, supervising 12/12 successful runs (173m 31s, zero OOMs).
- Rufet Dosteliyev (20.0%):** Built multi-seed aggregation engine in src/utlis/export\_latex.py and src/utlis/logger.py generating LaTeX tables directly from sweep\_summary.json with zero manual transcription. Authored visualization scripts for attention heatmaps and entropy trajectories, and compiled the IEEE conference manuscript (paper/main.pdf) and slide deck (presentation/slides.pdf).

## 4. Academic Integrity Attestation & Sign-Off

We certify that this report accurately reflects the collaborative division of labor for our final project. All members actively reviewed, validated, and approved the codebase, empirical findings, manuscript, and oral defense deck.

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|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Shahin Alakparov<br>Core Architecture Lead<br>Status: Verified & Signed | Gulnisa Abdurahmanli<br>Data Pipeline Lead<br>Status: Verified & Signed | Narmina Ibrahimova<br>Metrics & Hooks Lead<br>Status: Verified & Signed           |
| Emil Ahmedli<br>Sweep Operations Lead<br>Status: Verified & Signed      | Rufet Dosteliyev<br>Results & Paper Lead<br>Status: Verified & Signed   | Course Instruction Team<br>DLE-AI-202 Evaluation<br>Status: Submitted for Grading |